

Data Engineering Assignment 1

Part 3 – Apache Pig

Objective

Process an e-commerce review dataset using Apache Pig by loading the dataset from HDFS, filtering ratings below 1.0 or null values, grouping records by product category, and computing the average rating for each category.

Environment and HDFS Setup

Ubuntu 22.04.5 LTS, Hadoop 3.3.6, Apache Pig 0.17.0, Java 17, and HDFS were used. The review dataset was stored in HDFS at `/user/aiml/reviews/reviews.csv`.

Review Dataset

Review ID	Product Category	Rating
1	Electronics	4.5
2	Electronics	0.5
3	Electronics	4.0
4	Clothing	3.5
5	Clothing	0.8
6	Clothing	4.2
7	Books	4.8
8	Books	0.3
9	Books	4.0
10	Home	3.9
11	Home	
12	Home	4.1

Pig Latin Implementation

```
reviews = LOAD '/user/aiml/reviews/reviews.csv'
          USING PigStorage(',')
          AS (review_id:int, product_category:chararray, rating:double);

filtered_reviews = FILTER reviews BY rating IS NOT NULL AND rating >= 1.0;

grouped_reviews = GROUP filtered_reviews BY product_category;

average_ratings = FOREACH grouped_reviews
                  GENERATE group AS product_category,
                           AVG(filtered_reviews.rating) AS average_rating;

DUMP average_ratings;
```

Execution Command

```
pig -x mapreduce /home/aiml/Data-Engineering-Assignment-1/reviews.pig
```

Result

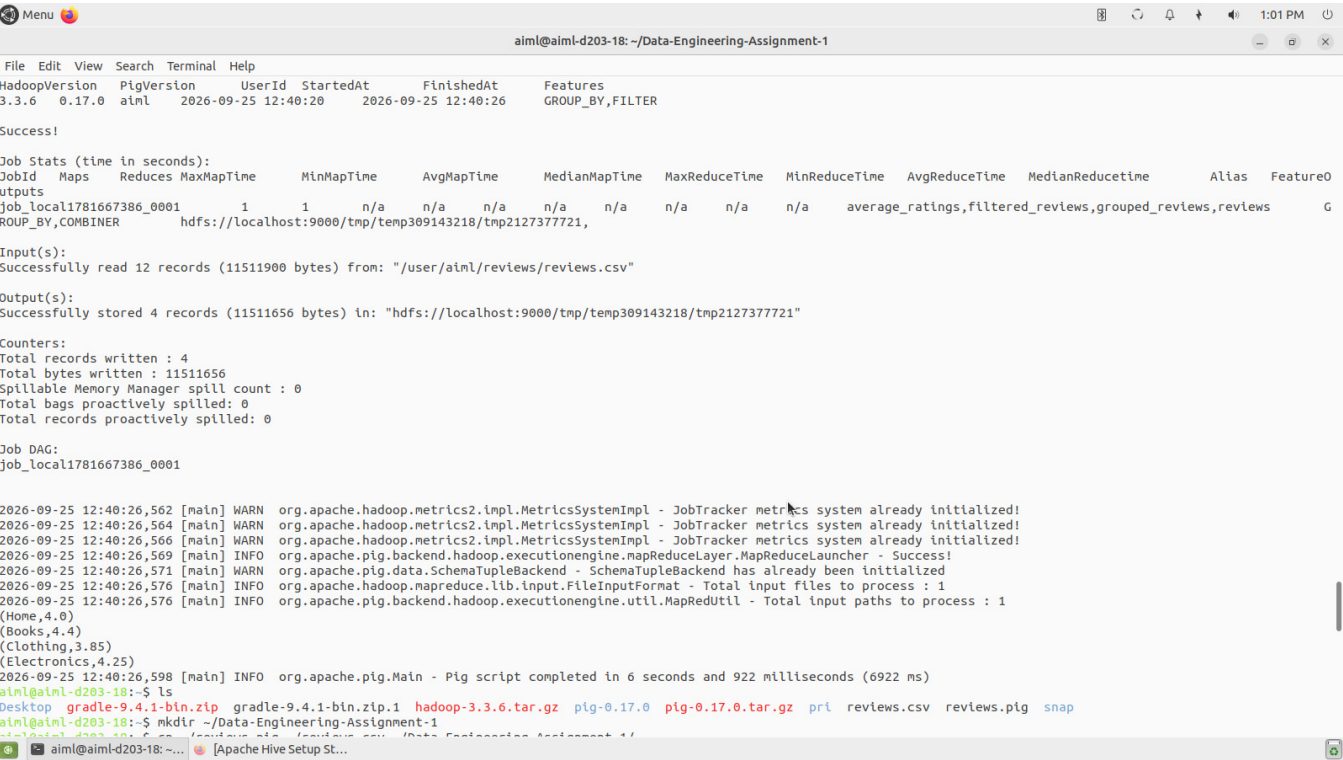
Product Category	Average Rating
Home	4.0
Books	4.4
Clothing	3.85
Electronics	4.25

(Home, 4.0)
(Books, 4.4)
(Clothing, 3.85)
(Electronics, 4.25)

Terminal Execution Screenshots

The following screenshots show the successful Apache Pig execution and the resulting category-wise average ratings.

Screenshot 1



Screenshot 2

```
aiml@aiml-d203-18: ~/Data-Engineering-Assignment-1
File Edit View Search Terminal Help
2026-09-25 12:40:26,540 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2026-09-25 12:40:26,541 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-25 12:40:26,559 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-25 12:40:26,561 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
3.3.6  0.17.0  aiml  2026-09-25 12:40:20  2026-09-25 12:40:26  GROUP_BY,FILTER

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReducetime  Alias  Feature0
utputs
job_local1781667386_0001  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  average_ratings,filtered_reviews,grouped_reviews,reviews  G
ROUP_BY,COMBINER  hdfs://localhost:9000/tmp/temp309143218/tmp2127377721,

Input(s):
Successfully read 12 records (11511900 bytes) from: "/user/aiml/reviews/reviews.csv"

Output(s):
Successfully stored 4 records (11511656 bytes) in: "hdfs://localhost:9000/tmp/temp309143218/tmp2127377721"

Counters:
Total records written : 4
Total bytes written : 11511656
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1781667386_0001

2026-09-25 12:40:26,562 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-25 12:40:26,564 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-25 12:40:26,566 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-25 12:40:26,569 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-25 12:40:26,571 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-09-25 12:40:26,576 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-25 12:40:26,576 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(Home,4.0)
(Books,4.4)
(Clothing,3.85)
(Electronics,4.25)
```

Screenshot 3

```
2026-09-25 12:40:26,576 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(Home,4.0)
(Books,4.4)
(Clothing,3.85)
(Electronics,4.25)
2026-09-25 12:40:26,598 [main] INFO org.apache.pig.Main - Pig script completed in 6 seconds and 922 milliseconds (6922 ms)
```

Discussion Questions

1. Why is **FILTER** used?

FILTER removes reviews with ratings below 1.0 and null ratings so that only valid ratings are used.

2. Why is **GROUP** used?

GROUP combines valid reviews according to product category so that an average can be calculated for each category.

3. Why is **AVG()** used?

AVG() calculates the mean rating of the valid reviews within each product category.

4. Where is the input dataset stored?

The dataset is stored in HDFS at `/user/aiml/reviews/reviews.csv`.

5. What is the result?

The script produces four category-wise average ratings: Home 4.0, Books 4.4, Clothing 3.85, and Electronics 4.25.

Conclusion

The Apache Pig script successfully processed the review dataset stored in HDFS. Invalid ratings were filtered, valid reviews were grouped by product category, and average ratings were calculated successfully.